



## LIGHTING CONTROL & MONITORING HELPED AADITYA TO INTRODUCE SMART LED STREET LIGHTS

System Level Solutions had provided an IoT framework with innovative Hardware and Software solution to Aaditya for smart street lights.

### Industry

Lighting

### Requirements

- To improve efficiency,
- To reduce cost,
- To maintain and sustain the quality of the product
- To enrich lives
- To protect the environment

### Solution

Lighting control monitoring system

### Features

- Security - Authorized user log-in
- Asset Management
- Binding between NebuLink OLC and Sensor
- Real-time Monitoring and Controlling
- Dimming Control
- Scheduling
- Alarms
- Reports/Data History
- Remote Troubleshooting

### Outcome

- Significant energy saving
- Reduced maintenance cost
- Energy/Power monitoring
- Efficient addressing street light outages and maintenance operations
- Street light ON/OFF schedule
- Reduction of CO2 emissions



### Summary

Aaditya Lighting was looking for solutions to make their street lights smart. SLS, as a solution partner, offered solutions with Hardware and software for lighting control and monitoring. This, in turn, helped Aaditya to offer lighting solution which reduced electricity consumption as well as improved public safety and satisfaction.

### Client Profile

Aaditya Lighting Private Limited, based in Surat, is an Industry leader in LED manufacturing with a wide range of LED home automation solutions. Using high-end technology and smart innovation techniques, the brand has now established itself as one of the leading energy efficient lighting solution brands in India.

### Business Requirements

Aaditya R&D has always been at the forefront of the industry to leverage the latest technologies, to improve efficiency, reduce cost, maintain and sustain the quality of the product, to enrich lives and to protect the environment. In sync with these objectives, they decided to make the street lights smart for reducing energy consumption as well as human labour. Therefore, Aaditya Lighting was in search of a suitable solution to centralized monitoring system using which their clients can control the lights, check faults and health of the lights.

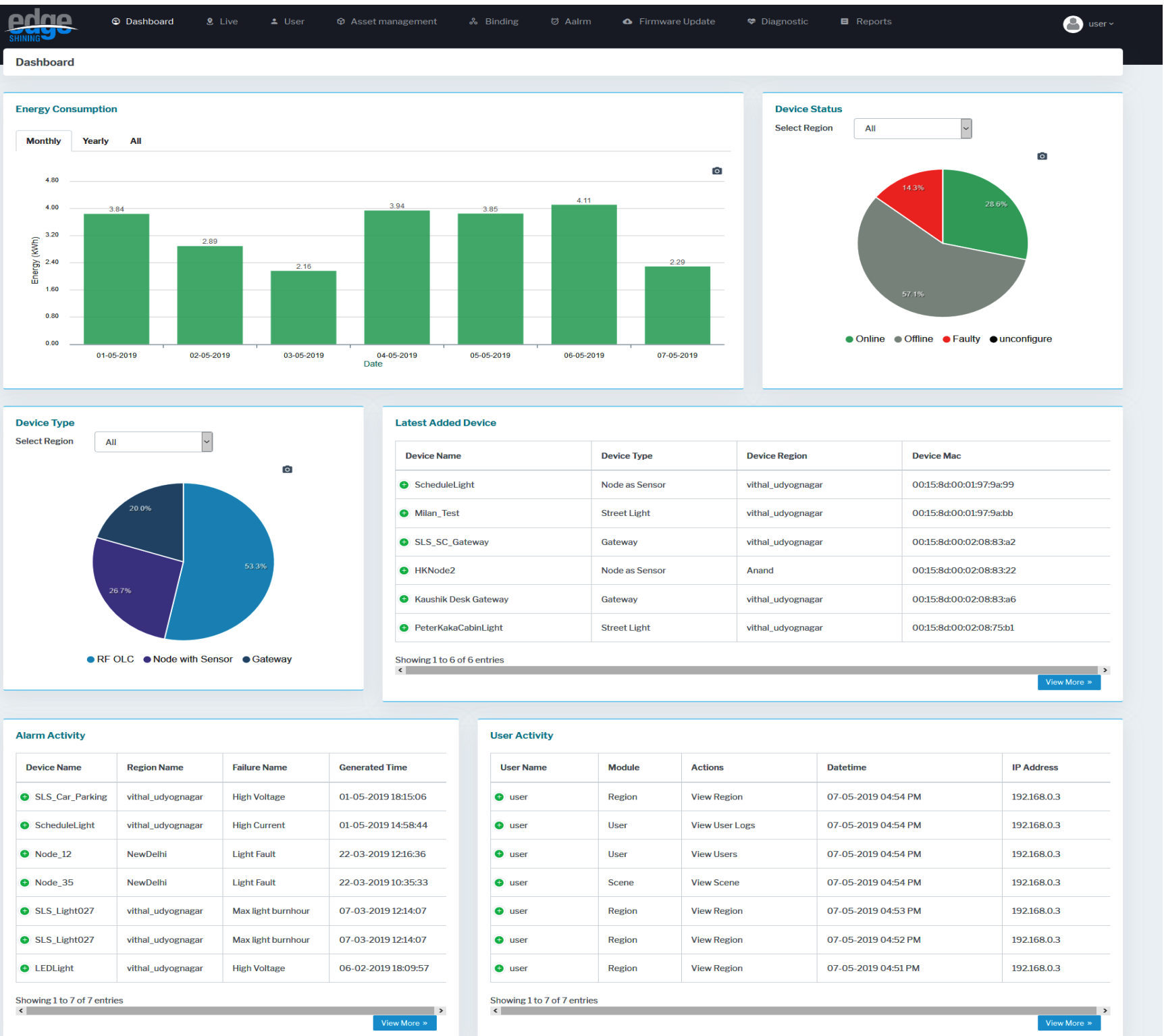
## The Solution

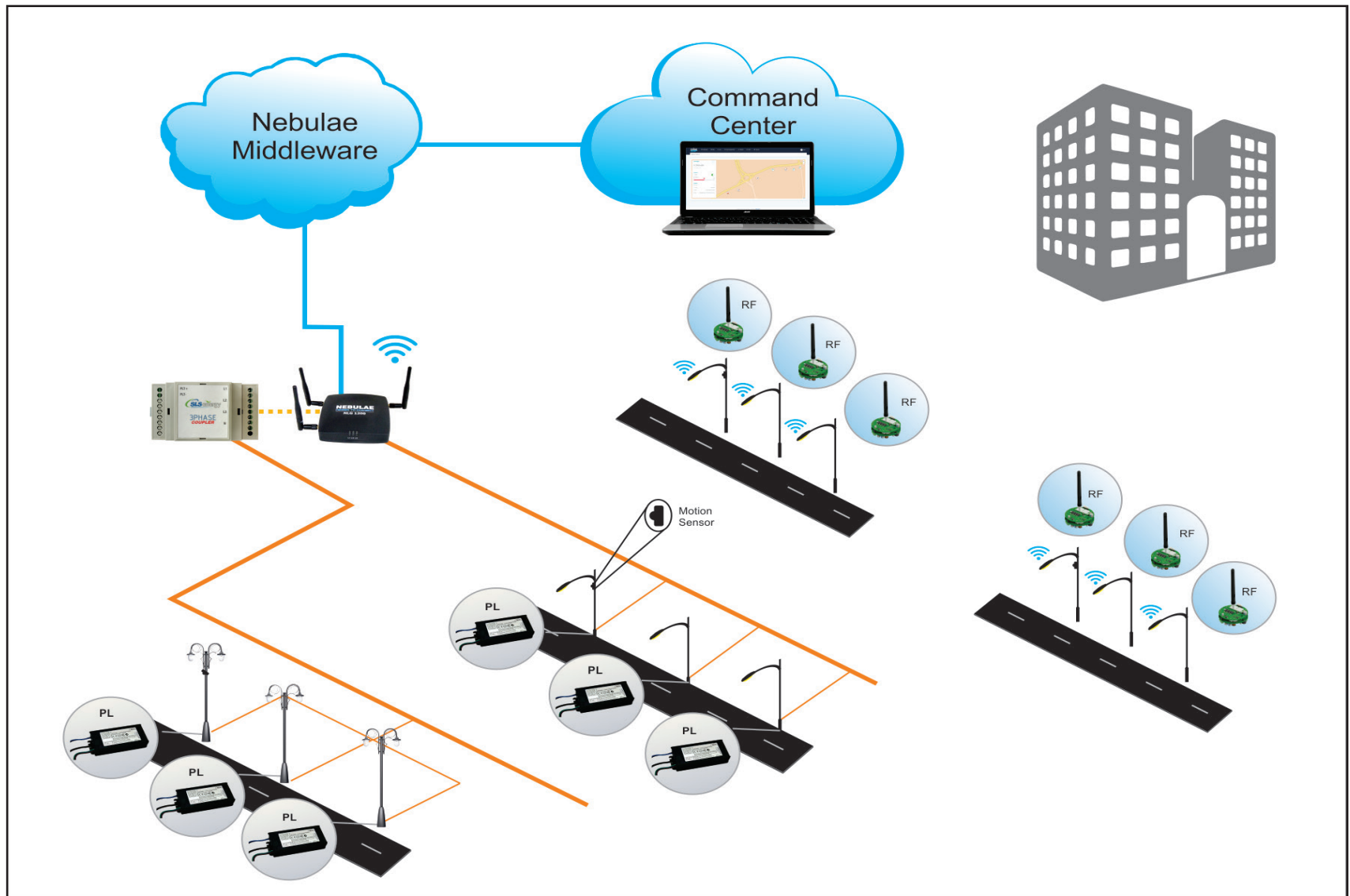
Aaditya Lighting selected SLS as the solution partner considering SLS's domain and technical expertise. SLS had already developed similar concept in other customer product in the same domain using its Nebulae framework and cloud-based solution. Moreover, SLS also got a competitive advantage because of its expertise to provide RF based Gateway, Nodes, Motion sensor and centralized portal to control the devices.

The whole solution is powered by our Nebulae framework along with our own cloud platform.

SLS proposed its IoT platform- Nebulae with RF Nodes and Gateways, where RF based Nodes are capable of communicating with RF based Gateways along with the motion sensor. These RF based Gateways have GSM support to send and retrieve data thorough cloud which can be monitored by centralized monitoring system. SLS along with hardware designing solution, provided software portal including architecture and planning.

The portal provides the summarize details of street light like energy consumption, device status, alarm activity, user activity etc. as shown in below image.





## Outcome

SLS has implemented solutions successfully meeting all required deadlines and expectations of Aaditya and will continue to develop further.

Nebulae powered RF based gateway and nodes are used in Aaditya's Smart LED street light solution - EdgeShining. The solution is cost effective and time efficient as after using this solution the users were able to reduce electricity consumption and man-hours requirement for light control. This in turn, improved public safety and satisfaction due to

reduced lamp downtime, less light pollution, and better overall lighting quality.

"We are very happy with the "Edge Shining" Street Light solution powered by SLS. This solution uses Nebulae IoT framework which is beyond imagination. It will help us to get more business in smart street light solution over India and ride smart city campaign in India."

**Nirav Dhamelia, MD and Founder, Aaditya Lighting**

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